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SUPPLEMENT TO "KERNEL ESTIMATORS OF ASYMPTOTIC VARIANCE FOR ADAPTIVE MARKOV CHAIN MONTE CARLO"

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This is a supplement to the paper "Kernel estimators of asymptotic variance for adaptive Markov Chain Monte Carlo" and contains the proofs to Theorems 4.1-4.3. For improved readability, we recall the theorems and their assumptions.

1. Statement of the theorems.

A1 *For each $\theta \in \Theta$, P_θ is phi-irreducible, aperiodic with invariant distribution π. There exists a measurable function $V: \mathsf{X} \to [1, \infty)$ with $\int V(x)\bar{\mu}(dx, d\theta) < \infty$ such that for any $\beta \in (0, 1]$, there exist $\rho \in (0, 1)$, $C \in (0, \infty)$ such that for any $(x, \theta) \in \mathsf{X} \times \Theta$,*

$$\|P_\theta^n(x, \cdot) - \pi(\cdot)\|_{V^\beta} \le C\rho^n \; V^\beta(x), \quad n \ge 0. \tag{1.1}$$

Furthermore, there exist constants $b \in (0, \infty), \lambda \in (0, 1)$ such that for any $(x, \theta) \in \mathsf{X} \times \Theta$,

$$P_\theta V(x) \le \lambda V(x) + b. \tag{1.2}$$

A2 *There exists $\eta \in [0, 1/2)$ and a non-increasing sequence of positive numbers $\{\gamma_n, \; n \ge 1\}$, $\gamma_n = O\left(n^{-\alpha}\right)$, $\alpha > 1/2$, such that for any $\beta \in [0, 1]$, there exists a finite constant C such that*

$$D_\beta\left(\theta_{n-1}, \theta_n\right) \le C\gamma_n V^\eta(X_n), \quad \mathbb{P} - a.s. \tag{1.3}$$

Let $w : \mathbb{R} \to \mathbb{R}$ be a function with support $[-1, 1]$ ($w(x) = 0$ for $|x| \ge 1$). We assume that w satisfies the following.

A3 The function w is even ($w(-x) = w(x)$) and $w(0) = 1$. Moreover, the restriction $w: \; [0, 1] \to \mathbb{R}$ is twice continuously differentiable.

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Let $\{b_n,\ n \geq 1\}$ be a nonincreasing sequence of positive numbers such that

$$(1.4)\qquad b_n^{-1} = O\left(n^{1/2}\right), \quad \text{and} \quad |b_n - b_{n-1}| = O\left(b_n n^{-1}\right), \ \text{ as } n \to \infty.$$

We consider the class of kernel estimator of the form

$$(1.5)\qquad \Gamma_n^2(h) = \sum_{k=-n}^{n} w(kb_n)\gamma_n(k) = \sum_{k=-b_n^{-1}+1}^{b_n^{-1}-1} w(kb_n)\gamma_n(k).$$

THEOREM 1.1. *Assume A1-A3. Let $\beta \in (0, 1/2-\eta)$ and $h \in \mathcal{L}_{V^\beta}$, where η is as in A2. Then*

$$(1.6)\qquad \Gamma_n^2(h) = \frac{1}{n}\sum_{k=1}^{n} G^2_{\theta_{k-1}}\left(X_{k-1}, X_k\right) + Q_n + D_n + \epsilon_n, \ \ n \geq 1.$$

The random process $\{(Q_n, D_n, \epsilon_n),\ n \geq 1\}$ is such that for any $p > 1$ such that $2p(\beta+\eta) \leq 1$, there exists a finite constant C such that

$$(1.7)\qquad \mathbb{E}\left(|Q_n|^p\right) \leq C\left(b_n + n^{-\alpha}b_n^{-1+\alpha} + n^{-1+\frac{1}{2}\vee\frac{1}{p}}b_n^{-1/2}\right)^p,$$
$$\mathbb{E}\left(|D_n|^p\right) \leq Cb_n^p \quad \text{and} \quad \mathbb{E}\left(|\epsilon_n|^p\right) \leq C\left(n^{-1}b_n^{-1}\right)^p.$$

In particular if $\lim_{n\to\infty} n^{-1+\frac{1}{2}\vee\frac{1}{p}}b_n^{-1/2} = 0$ then $\Gamma_n^2(h) - \frac{1}{n}\sum_{k=1}^{n} G^2_{\theta_{k-1}}\left(X_{k-1}, X_k\right)$ converges to zero in L^p.

THEOREM 1.2. *Assume A1-A3 with $\eta < 1/4$, where η is as in A2. Let $\beta \in (0, 1/4-\eta)$ and $h \in \mathcal{L}_{V^\beta}$. Suppose that $b_n \propto n^{-\delta}$ where $\delta \in (2(\beta+\eta), 1/2)$. Then*

$$\lim_{n\to\infty}\left(\Gamma_n^2(h) - \frac{1}{n}\sum_{k=1}^{n} G^2_{\theta_{k-1}}\left(X_{k-1}, X_k\right)\right) = 0,$$

almost surely.

A4 The function w is even ($w(-x) = w(x)$) and $w(0) = 1$. Moreover, the restriction $w:\ [0, 1] \to \mathbb{R}$ is $(r+1)$-times continuously differentiable for some $r \geq 2$.

THEOREM 1.3. *Assume A1-A2 and A4. Let $\beta \in (0, 1/2 - \eta)$ and $h \in \mathcal{L}_{V^\beta}$, where η is as in A2. Let $p > 1$ be such that $2p(\beta + \eta) \le 1$. Suppose in addition that*

$$\sum_{n\ge 1}\left(n^{-1}b_n^{-1}\right)^p < \infty, \quad \sum_{n\ge 1}\left(n^{-2}b_n^{-1}\right)^{1\wedge(p/2)} < \infty, \qquad (1.8)$$

$$\sum_{n\ge 1} n^{-2+\frac{1}{2}\vee\frac{1}{p}} b_n^{-1/2} < \infty \quad \text{and} \quad \sum_{n\ge 1} b_n^{(r-1)p} < \infty.$$

Then the conclusion of Theorem 1.2 holds.

2. Proofs. Throughout the proof, we shall use C to denote a generic constant. At multiple occasions we make use of the Kronecker Lemma and the Toeplitz Lemma. We refer the reader to [3] Section 2.6 for a statement and proof of these lemmata. Also we shall use the following characterization of almost sure convergence: a sequence $\{X_n,\ n \ge 0\}$ converges almost surely to zero if and only if for any $\delta > 0$, $\lim_{n\to\infty} \mathbb{P}\left(\sup_{m\ge n}|X_m| > \delta\right) = 0$. In particular if for any $\delta > 0$, $\sum_n \mathbb{P}(|X_n| > \delta) < \infty$ then $\{X_n,\ n \ge 0\}$ converges almost surely to zero.

We shall routinely use the following martingale inequality. Let $\{D_i, \mathcal{F}_i,\ i \ge 1\}$ be a martingale difference sequence. For any $p > 1$,

$$\mathbb{E}\left(\left|\sum_{i=1}^n D_i\right|^p\right) \le C\left\{\sum_{i=1}^n \mathbb{E}^{1\wedge(2/p)}\left(|D_i|^p\right)\right\}^{1\vee(p/2)}, \qquad (2.1)$$

where C can be taken as $C = \left(18pq^{1/2}\right)^p$, $p^{-1} + q^{-1} = 1$.

We introduce the resolvent functions

$$g_\theta(x) \stackrel{\text{def}}{=} \sum_{j\ge 0} \bar{P}_\theta^j h(x),$$

where $\bar{P}_\theta \stackrel{\text{def}}{=} P_\theta - \pi$. The dependence of g_θ on h is omitted for convenience in the notation. We also define $G_\theta(x, y) = g_\theta(y) - P_\theta g_\theta(x)$, where $P_\theta g_\theta(x) \stackrel{\text{def}}{=} \int P_\theta(x, dz) g_\theta(z)$. The function g_θ satisfies the so-called *Poisson equation*

$$h(x) = g_\theta(x) - \bar{P}_\theta g_\theta(x). \qquad (2.2)$$

We also notice that for any $q \in [1, \beta^{-1}]$ Lemma A.1 (a)-(b) implies that

$$\sup_{k\ge 1} \mathbb{E}\left(|G_{\theta_{k-1}}(X_{k-1}, X_k)|^q\right) < \infty. \qquad (2.3)$$

Finally, the following consequence of A3 will be routinely used. See Appendix B for more details. There exists a finite constant c such that for any $n \geq 1$,

$$\sum_{\ell=-n}^{n} |w(\ell b_n)| \leq c b_n^{-1}, \quad \sum_{\ell=-n}^{n} |w((\ell-1)b_n) - w(\ell b_n)| \leq c, \quad \text{and} \quad \sum_{\ell=-n}^{n} |A_{n,\ell-1}| \leq c b_n, \tag{2.4}$$

where $A_{n,\ell} = 2w(\ell b_n) - w((\ell-1)b_n) - w((\ell+1)b_n)$.

Lemma 2.1. *Assume A1-A3. Let $\beta \in [0, 1/2-\eta)$ and $h \in \mathcal{L}_{V^\beta}$. We have the decomposition*

$$\Gamma_n^2(h) - \frac{1}{n}\sum_{k=1}^{n} G^2_{\theta_{k-1}}(X_{k-1}, X_k) = \sum_{i=1}^{5} D_n^{(i)} + \sum_{i=1}^{4} Q_n^{(i)} + \epsilon_n, \quad n \geq 1. \tag{2.5}$$

where

$$D_n^{(1)} = 2n^{-1}(1 - w(b_n)) \sum_{j=1}^{n} P_{\theta_{j-1}} g_{\theta_{j-1}}(X_{j-1}) G_j,$$

$$D_n^{(2)} = 2n^{-1} \sum_{j=1}^{n} \left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right) G_j$$

$$D_n^{(3)} = 2n^{-1}(1 - w(b_n)) \sum_{j=1}^{n} \left(P_{\theta_{j-1}} g_{\theta_{j-1}}(X_{j-1})\right)^2,$$

$$D_n^{(4)} = n^{-1} \sum_{j=1}^{n} \left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right)^2$$

$$D_n^{(5)} = 2n^{-1}(1 - w(b_n)) \sum_{j=1}^{n} P_{\theta_{j-1}} g_{\theta_{j-1}}(X_{j-1}) \left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right),$$

$$Q_n^{(1)} = 2n^{-1} \sum_{j=2}^{n} Z_{n,j}^{(1)} G_j, \quad Q_n^{(2)} = 4n^{-1} \sum_{j=2}^{n} \left(Z_{n,j}^{(2)} + Z_{n,j}^{(3)}\right) G_j,$$

$$Q_n^{(3)} = 2n^{-1} \sum_{j=2}^{n} \left(Z_{n,j}^{(3)} + 2Z_{n,j}^{(2)}\right)\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right),$$

$$Q_n^{(4)} = n^{-1} \sum_{j=1}^{n-1} Z_{n,j}^{(4)} P_{\theta_j} g_{\theta_j}(X_j), \quad \textit{where} \quad Z_{n,j}^{(1)} = \sum_{\ell=1}^{j-1} w((j-\ell)b_n) G_\ell,$$

$$Z^{(2)}_{n,j} = \sum_{\ell=1}^{j-1} \left(w\left((j-\ell)b_n\right) - w\left((j-\ell+1)b_n\right)\right) P_{\theta_{\ell-1}} g_{\theta_{\ell-1}}(X_{\ell-1})$$

$$Z^{(3)}_{n,j} = \sum_{\ell=1}^{j-1} w\left((j-\ell)b_n\right)\left(g_{\theta_{\ell-1}}(X_\ell) - g_{\theta_\ell}(X_\ell)\right) \quad \textit{and} \quad Z^{(4)}_{n,j} = \sum_{\ell=0}^{j-1} A_{n,j-\ell} P_{\theta_\ell} g_{\theta_\ell}(X_\ell).$$

where

$$A_{n,\ell} = 2w\left(\ell b_n\right) - w\left((\ell-1)b_n\right) - w\left((\ell+1)b_n\right).$$

Furthermore, for any $p > 1$ *such that* $2p(\beta+\eta) \le 1$.

$$\mathbb{E}\left(|\epsilon_n|^p\right) \le C\left(n^{-1}b_n^{-1}\right)^p. \tag{2.6}$$

Proof. Given the expression of $\gamma_n(k)$, we can assume without any loss of generality that $\pi(h) = 0$. Let us introduce the partial sum $S_k(j) = \sum_{\ell=j+1}^{j+k} h(X_\ell)$. Then we can rewrite $\gamma_n(k)$ ($|k| < n$) as

$$\gamma_n(k) = \frac{1}{n}\sum_{j=1}^{n-k} h(X_j)h(X_{j+k}) + R^{(1)}_n(k),$$

where

$$R^{(1)}_n(k) = -\frac{1}{n^2} S_n(0)\left(S_{n-k}(0) + S_{n-k}(k)\right) + \frac{n-k}{n^3} S_n(0)^2.$$

We introduce the periodogram

$$I_n(u) = \frac{1}{2\pi n}\left|\sum_{k=1}^{n} e^{-iku} h(X_k)\right|^2 = \frac{1}{2\pi n}\left(\sum_{k=1}^{n} e^{-iku} h(X_k)\right)\left(\sum_{k=1}^{n} e^{iku} h(X_k)\right),$$

where i denotes the imaginary unit number. Notice that $I_n(-u) = I_n(u)$. It is well know and easy to check using the orthonormality of the trigonometric family $\{e^{iku},\ k \in \mathbb{Z}\}$ ($(2\pi)^{-1}\int_{-\pi}^{\pi} e^{iku}e^{-ik'u}du = \delta_{\{k=k'\}}$) that the term $\frac{1}{n}\sum_{j=1}^{n-k} h(X_j)h(X_{j+k})$ can be re-written as

$$\frac{1}{n}\sum_{j=1}^{n-k} h(X_j)h(X_{j+k}) = \int_{-\pi}^{\pi} e^{iku} I_n(u)du.$$

This offers another representation of $\Gamma^2_n(h)$ as

$$\Gamma^2_n(h) = \sum_{k=-n}^{n} w(kb_n)\int_{-\pi}^{\pi} e^{iuk} I_n(u)du \ + \ \sum_{k=-n}^{n} w(kb_n) R^{(1)}_n(k). \tag{2.7}$$

The right-most term in (2.7) is negligible. Indeed using (c) of Lemma A.1 and since $p > 1$ we have

$$\mathbb{E}\left(|R_n^{(1)}(k)|^p\right) \leq n^{-2p}\left\{\mathbb{E}^{1/p}\left(|S_n(0)|^{2p}\right)\right.$$
$$\left.+\mathbb{E}^{1/2p}\left(|S_n(0)|^{2p}\right) \times \left[\mathbb{E}^{1/2p}\left(|S_{n-k}(0)|^{2p}\right) + \mathbb{E}^{1/2p}\left(|S_{n-k}(k)|^{2p}\right)\right]\right\}^p \leq Cn^{-p},$$

for some finite constant C that does not depend on k nor n. Using this and the fact that w is bounded with support on $[-1, 1]$, we get

$$\mathbb{E}\left(\left|\sum_{k=-n}^{n} w(kb_n)R_n^{(1)}(k)\right|^p\right) \leq C\left(n^{-1}\sum_{k=-n}^{n}|w(kb_n)|\right)^p \leq C\left(n^{-1}b_n^{-1}\right)^p. \tag{2.8}$$

Now, we use the Poisson equation (2.2) to decompose

$$\sum_{k=1}^{n} e^{-iku}h(X_k) = \sum_{k=1}^{n} e^{-iku}G_{\theta_{k-1}}(X_{k-1}, X_k) + T_n^{(1)}(u),$$

where

$$T_n^{(1)}(u) = P_{\theta_0}g_{\theta_0}(X_0) - e^{-inu}P_{\theta_n}g_{\theta_n}(X_n) \tag{2.9}$$
$$+\sum_{k=1}^{n}\left(e^{-iu}-1\right)e^{-iu(k-1)}P_{\theta_{k-1}}g_{\theta_{k-1}}(X_{k-1}) + \sum_{k=1}^{n} e^{-iuk}\left(g_{\theta_{k-1}}(X_k) - g_{\theta_k}(X_k)\right).$$

We shall write G_k instead $G_{\theta_{k-1}}(X_{k-1}, X_k)$. It follows that

$$I_n(u) = \frac{1}{2\pi n}\left|\sum_{k=1}^{n} e^{-iku}G_k\right|^2 + \frac{1}{2\pi n}\left|T_n^{(1)}(u)\right|^2 + \frac{1}{2\pi n}\overline{\left(T_n^{(1)}(u)\right)}\sum_{k=1}^{n} e^{-iku}G_k + \frac{1}{2\pi n}T_n^{(1)}(u)\sum_{k=1}^{n} e^{iku}G_k, \tag{2.10}$$

and where for a complex number x, $\bar{x}$ denotes its conjugate. Now notice that

$$\left|\sum_{j=1}^{n} e^{-iju}G_k\right|^2 = \sum_{1\leq j,j'\leq n} e^{i(j'-j)u}G_{j'}G_j$$

and $(2\pi)^{-1}\int_{-\pi}^{\pi} e^{iku}e^{i(j'-j)u}du = \delta_{\{j-j'\}}(k)$. Thus

$$\frac{1}{2\pi n}\sum_{k=-n}^{n} w(kb_n)\int_{-\pi}^{\pi} e^{iku}\left|\sum_{j=1}^{n} e^{-iju}G_j\right|^2 du$$
$$= n^{-1}\sum_{1\le j,j'\le n} w\left((j'-j)b_n\right)G_jG_{j'} = n^{-1}\sum_{j=1}^{n} G_j^2 + Q_n^{(1)}. \tag{2.11}$$

where

$$Q_n^{(1)} = 2n^{-1}\sum_{j=2}^{n} Z_{n,j}^{(1)}G_j, \qquad Z_{n,j}^{(1)} = \sum_{\ell=1}^{j-1} w\left((j-\ell)b_n\right)G_\ell.$$

We move to the last two terms of (2.10). Given $T_n^{(1)}(u)$ in (2.9), we have

$$\left(\sum_{j=1}^{n} e^{iju}G_j\right)T_n^{(1)}(u) = P_{\theta_0}g_{\theta_0}(X_0)\sum_{j=1}^{n} e^{iju}G_j - P_{\theta_n}g_{\theta_n}(X_n)\sum_{j=1}^{n} e^{-i(n-j)u}G_j$$
$$+\sum_{j'=1}^{n}\sum_{j=1}^{n}\left(e^{-iu}-1\right)e^{i(j'-j+1)u}G_{j'}P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})$$
$$+\sum_{j'=1}^{n}\sum_{j=1}^{n} e^{i(j'-j)u}G_{j'}\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right).$$

Therefore

$$\sum_{k=-n}^{n} w(kb_n)\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{iku}\left(\sum_{j=1}^{n} G_je^{iju}\right)T_n^{(1)}(u)du = P_{\theta_0}g_{\theta_0}(X_0)\sum_{j=1}^{n} w(jb_n)G_j$$
$$-P_{\theta_n}g_{\theta_n}(X_n)\sum_{j=1}^{n} w((n-j)b_n)G_j$$
$$+\sum_{j'=1}^{n}\sum_{j=1}^{n}\left(w\left((j'-j)b_n\right) - w\left((j'-j+1)b_n\right)\right)G_{j'}P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})$$
$$+\sum_{j'=1}^{n}\sum_{j=1}^{n} w\left((j'-j)b_n\right)G_{j'}\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right).$$

It follows that

$$\sum_{k=-n}^{n} w(kb_n)\frac{1}{2\pi n}\int_{-\pi}^{\pi} e^{iku}\left\{\left(\sum_{j=1}^{n} G_je^{iju}\right)T_n^{(1)}(u) + \left(\sum_{j=1}^{n} G_je^{-iju}\right)\overline{T_n^{(1)}(u)}\right\}du$$
$$= D_n^{(1)} + D_n^{(2)} + Q_n^{(2)} + U_n^{(1)}, \tag{2.12}$$

where

$$D_n^{(1)} = 2n^{-1}\left(1 - w(b_n)\right)\sum_{j=1}^{n} P_{\theta_{j-1}} g_{\theta_{j-1}}(X_{j-1}) G_j,$$

$$D_n^{(2)} = 2n^{-1}\sum_{j=1}^{n}\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right) G_j.$$

and $Q_n^{(2)}$ is the quadratic form

$$Q_n^{(2)} = 4n^{-1}\sum_{j=2}^{n}\left(Z_{n,j}^{(2)} + Z_{n,j}^{(3)}\right) G_j,$$

$$Z_{n,j}^{(2)} = \sum_{\ell=1}^{j-1}\left(w\left((j-\ell)b_n\right) - w\left((j-\ell+1)b_n\right)\right) P_{\theta_{\ell-1}} g_{\theta_{\ell-1}}(X_{\ell-1}) \tag{2.13}$$

$$Z_{n,j}^{(3)} = \sum_{\ell=1}^{j-1} w\left((j-\ell)b_n\right)\left(g_{\theta_{\ell-1}}(X_\ell) - g_{\theta_\ell}(X_\ell)\right).$$

with a remainder

$$U_n^{(1)} = 2n^{-1}\sum_{j=1}^{n}\left\{w(jb_n) P_{\theta_0} g_{\theta_0}(X_0) - 2w((n-j)b_n) P_{\theta_n} g_{\theta_n}(X_n)\right\} G_j.$$

Since $\{(G_j, \mathcal{F}_j),\ j \geq 1\}$ is a martingale difference, one can easily show that $U_n^{(1)}$ satisfies

$$\mathbb{E}\left(\left|U_n^{(1)}\right|^p\right) \leq C\left(n^{-1} b_n^{-1/2}\right)^p. \tag{2.14}$$

We now turn to the term $(2\pi n)^{-1}\left|T_n^{(1)}(u)\right|^2$ in (2.10). We have

$$\begin{aligned}\left|T^{(1)}(u)\right|^2 &= \left(P_{\theta_0} g_{\theta_0}(X_0)\right)^2 + \left(P_{\theta_n} g_{\theta_n}(X_n)\right)^2 \\ &+ \sum_{1\leq j', j\leq n}\left(1 - e^{iu} - e^{-iu} + 1\right) e^{i(j'-j)u} P_{\theta_{j'-1}} g_{\theta_{j'-1}}(X_{j'-1}) P_{\theta_{j-1}} g_{\theta_{j-1}}(X_{j-1}) \\ &+ \sum_{1\leq j', j\leq n} e^{i(j'-j)u}\left(g_{\theta_{j'-1}}(X_{j'}) - g_{\theta'_j}(X_{j'})\right)\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right) + A + \overline{A},\end{aligned}$$

where

$$
\begin{aligned}
A = &-e^{inu}P_{\theta_0}g_{\theta_0}(X_0)P_{\theta_n}g_{\theta_n}(X_n)\\
&+(e^{iu}-1)\sum_{j=1}^{n}e^{iu(j-1)}P_{\theta_0}g_{\theta_0}(X_0)P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})\\
&+\sum_{j=1}^{n}e^{iuj}P_{\theta_0}g_{\theta_0}(X_0)\left(g_{\theta_{j-1}}(X_j)-g_{\theta_j}(X_j)\right)\\
&-(e^{iu}-1)\sum_{j=1}^{n}e^{-iu(n-j+1)}P_{\theta_n}g_{\theta_n}(X_n)P_{\theta_j}g_{\theta_j}(X_j)\\
&-\sum_{j=1}^{n}e^{-iu(n-k)}P_{\theta_n}g_{\theta_n}(X_n)\left(g_{\theta_{j-1}}(X_j)-g_{\theta_j}(X_j)\right)\\
&+\sum_{1\leq j',j\leq n}(e^{-iu}-1)e^{-iu(j'-j-1)}P_{\theta_{j'-1}}g_{\theta_{j'-1}}(X_{j'})\left(g_{\theta_{j-1}}(X_j)-g_{\theta_j}(X_j)\right).
\end{aligned}
$$

Therefore

$$
\begin{aligned}
\sum_{k=-n}^{n}w(kb_n)\frac{1}{2\pi}\int_{-\pi}^{\pi}e^{iku}\left|T_n^{(1)}(u)\right|^2du = &\left(P_{\theta_0}g_{\theta_0}(X_0)\right)^2+\left(P_{\theta_n}g_{\theta_n}(X_n)\right)^2\\
&+\sum_{1\leq j',j\leq n}A_{n,j'-j}P_{\theta_{j'-1}}g_{\theta_{j'-1}}(X_{j'-1})P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})\\
&+\sum_{1\leq j',j\leq n}w\left((j'-j)b_n\right)\left(g_{\theta_{j'-1}}(X_{j'})-g_{\theta_j'}(X_{j'})\right)\left(g_{\theta_{j-1}}(X_j)-g_{\theta_j}(X_j)\right)\\
&-2w(nb_n)P_{\theta_0}g_{\theta_0}(X_0)P_{\theta_n}g_{\theta_n}(X_n)\\
&+2\sum_{j=1}^{n}\left(w(jb_n)-w((j-1)b_n)\right)P_{\theta_0}g_{\theta_0}(X_0)P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})\\
&+2\sum_{j=1}^{n}w(jb_n)P_{\theta_0}g_{\theta_0}(X_0)\left(g_{\theta_{j-1}}(X_j)-g_{\theta_j}(X_j)\right)\\
&-2\sum_{j=1}^{n}\left(w((n-j)b_n)-w((n-j+1)b_n)\right)P_{\theta_n}g_{\theta_n}(X_n)P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})\\
&-2\sum_{j=1}^{n}w((n-j)b_n)P_{\theta_n}g_{\theta_n}(X_n)\left(g_{\theta_{j-1}}(X_j)-g_{\theta_j}(X_j)\right)\\
&+2\sum_{1\leq j',j\leq n}\Delta w_{n,j'-j}P_{\theta_{j'-1}}g_{\theta_{j'-1}}(X_{j'-1})\left(g_{\theta_{j-1}}(X_j)-g_{\theta_j}(X_j)\right),
\end{aligned}
$$

where $A_{n,\ell}=2w(\ell b_n)-w((\ell-1)b_n)-w((\ell+1)b_n)$, and $\Delta w_{n,\ell}=w(\ell b_n)-$

$w((\ell-1)b_n)$. It follows that

$$\frac{1}{2\pi n}\sum_{k=-n}^{n} w(kb_n)\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{iku}\left|T_n^{(1)}(u)\right|^2 du \\ = D_n^{(3)} + D_n^{(4)} + D_n^{(5)} + Q_n^{(3)} + Q_n^{(4)} + U_n^{(2)}, \tag{2.15}$$

where $D_n^{(3)}, D_n^{(4)}, D_n^{(5)}$ regroups the diagonal terms

$$D_n^{(3)} = 2n^{-1}(1-w(b_n))\sum_{j=1}^{n}\left(P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})\right)^2,$$

$$D_n^{(4)} = n^{-1}\sum_{j=1}^{n}\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right)^2$$

and $$D_n^{(5)} = 2n^{-1}(1-w(b_n))\sum_{j=1}^{n} P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right)$$

$$Q_n^{(3)} = 2n^{-1}\sum_{j=2}^{n}\left(Z_{n,j}^{(3)} + 2Z_{n,j}^{(2)}\right)\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right),$$

where $Z_{n,j}^{(2)}$ and $Z_{n,j}^{(3)}$ are as given in (2.13). $Q_n^{(4)}$ is given by

$$Q_n^{(4)} = n^{-1}\sum_{j=1}^{n-1} Z_{n,j}^{(4)} P_{\theta_j}g_{\theta_j}(X_j), \quad Z_{n,j}^{(4)} = \sum_{\ell=0}^{j-1} A_{n,j-\ell}P_{\theta_\ell}g_{\theta_\ell}(X_\ell),$$

and

$$\begin{aligned} U_n^{(2)} = {} & n^{-1}\left(P_{\theta_0}g_{\theta_0}(X_0)\right)^2 + n^{-1}\left(P_{\theta_n}g_{\theta_n}(X_n)\right)^2 \\ & - 2n^{-1}w(nb_n)P_{\theta_0}g_{\theta_0}(X_0)P_{\theta_n}g_{\theta_n}(X_n) \\ & + 2n^{-1}\sum_{j=1}^{n}\left(w(jb_n) - w((j-1)b_n)\right)P_{\theta_0}g_{\theta_0}(X_0)P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1}) \\ & + 2n^{-1}\sum_{j=1}^{n} w(jb_n)P_{\theta_0}g_{\theta_0}(X_0)\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right) \\ & - 2n^{-1}\sum_{j=1}^{n}\left(w((n-j)b_n) - w((n-j+1)b_n)\right)P_{\theta_n}g_{\theta_n}(X_n)P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1}) \\ & - 2n^{-1}\sum_{j=1}^{n} w((n-j)b_n)P_{\theta_n}g_{\theta_n}(X_n)\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right) \end{aligned} \tag{2.16}$$

It is easy to check by using A3, Lemma A.1 and Proposition B.1 that

$$\mathbb{E}\left(\left|U_n^{(2)}\right|^p\right) \leq C\left(n^{-1}b_n^{-1}\right)^p. \tag{2.17}$$

Putting together (2.7, 2.10, 2.11,2.12,2.15) we get

$$\Gamma_n^2(h) - \frac{1}{n}\sum_{j=1}^n G_j^2 = \sum_{i=1}^5 D_n^{(i)} + \sum_{i=1}^4 Q_n^{(i)} + \epsilon_n, \tag{2.18}$$

where $\epsilon_n = \sum_{k=-n}^n w(kb_n)R_n^{(1)}(k) + U_n^{(1)} + U_n^{(2)}$. By (2.8,2.14 and 2.17) we have

$$\mathbb{E}\left(|\epsilon_n|^p\right) \leq C\left(n^{-1}b_n^{-1}\right)^p.$$

□

2.1. *Proof of Theorem 1.1.*

PROOF. We bound the p-th moment for each term on the rhs of (2.5). By A3, Lemma A.1 and (2.1)

$$\mathbb{E}\left(\left|n^{-1}\left(1-w(b_n)\right)\sum_{j=1}^n P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})G_j\right|^p\right)$$
$$\leq C\left(n^{-1}b_n\right)^p\left\{\sum_{j=1}^n \mathbb{E}^{1\wedge(2/p)}\left(\left|P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})G_j\right|^p\right)\right\}^{1\vee(p/2)} \leq Cn^{-p+1\vee\frac{p}{2}}b_n^p.$$

By Lemma A.1 and Minkowski's inequality

$$\mathbb{E}\left(\left|n^{-1}\sum_{j=1}^n\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right)G_j\right|^p\right) \leq Cn^{-\alpha p},$$

$$\mathbb{E}\left(\left|n^{-1}(1-w(b_n))\sum_{j=1}^n\left(P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})\right)^2\right|^p\right) \leq Cb_n^p,$$

$$\mathbb{E}\left(\left|n^{-1}\sum_{j=1}^n\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right)^2\right|^p\right) \leq Cn^{-2\alpha p}, \quad \text{and}$$

$$\mathbb{E}\left(\left|n^{-1}(1-w(b_n))\sum_{j=1}^n P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})\left(g_{\theta_{j-1}}(X_j) - g_{\theta_j}(X_j)\right)\right|^p\right) \leq Cb_n^p n^{-\alpha p}.$$

To sum up we have

$$\mathbb{E}\left(\left|\sum_{i=1}^{5} D_n^{(i)}\right|^p\right) \le C b_n^p.$$

$\{Z_{n,\ell}^{(1)},\ 1 \le \ell \le j-1\}$ is also a martingale for each n and using (2.1)

$$\mathbb{E}\left(\left|Z_{n,j}^{(1)}\right|^{2p}\right) \le C\left\{\sum_{\ell=1}^{j-1} \mathbb{E}^{1/p}\left(w^{2p}\left((j-\ell)b_n\right)|G_\ell|^{2p}\right)\right\}^p \le C\left\{\sum_{\ell=1}^{j-1} w^2\left((j-\ell)b_n\right)\right\}^p \le C b_n^{-p}. \tag{2.19}$$

In the last inequality, we use (2.4).

Since $\{\sum_{j=2}^k Z_{n,j}^{(1)} G_j, \mathcal{F}_j,\ 2 \le k \le n\}$ is a martingale for each n, it follows from (2.19) and (2.1) that

$$\mathbb{E}\left(\left|Q_n^{(1)}\right|^p\right) \le C n^{-p}\left\{\sum_{j=2}^{n} \mathbb{E}^{1\wedge(2/p)}\left(\left|Z_{n,j}^{(1)} G_j\right|^p\right)\right\}^{1\vee p/2} \le n^{-p+1\vee\frac{p}{2}} b_n^{-p/2}.$$

Using the bound (2.4) and Lemma A.1, we have

$$\mathbb{E}\left(\left|Z_{n,j}^{(2)}\right|^{2p}\right) \le C\left\{\sum_{\ell=1}^{j-1} \left|w\left((j-\ell)b_n\right) - w\left((j-\ell+1)b_n\right)\right| \mathbb{E}^{1/2p}\left(V^{2p\beta}(X_\ell)\right)\right\}^{2p} \le C\left\{\sum_{\ell=1}^{j-1} \left|w\left(\ell b_n\right) - w\left((\ell+1)b_n\right)\right|\right\}^{2p} \le C. \tag{2.20}$$

Similarly

$$\mathbb{E}\left(\left|Z_{n,j}^{(3)}\right|^{2p}\right) \le C\left\{\sum_{\ell=1}^{j-1} w\left((j-\ell)b_n\right)\gamma_\ell \mathbb{E}^{1/2p}\left(V^{2p(\eta+\beta)}(X_\ell)\right)\right\}^{2p} \le C\left\{\sum_{\ell=1}^{b_n^{-1}} \gamma_\ell\right\}^{2p} \le C b_n^{-2p(1-\alpha)}. \tag{2.21}$$

Since $Z_{n,j}^{(2)}$ and $Z_{n,j}^{(3)}$ are $\mathcal{F}_{j-1}$-measurable, $\{\sum_{j=2}^k \left(Z_{n,j}^{(2)} + Z_{n,j}^{(3)}\right) G_j, \mathcal{F}_j,\ 2 \le k \le n\}$ remains a martingale for each n. Using (2.20), (2.21) and (2.1) we thus obtain

$$\mathbb{E}\left(\left|Q_n^{(2)}\right|^p\right) \leq Cn^{-p}\left\{\sum_{j=2}^n \mathbb{E}^{1\wedge(2/p)}\left(\left|\left(Z_{n,j}^{(2)}+Z_{n,j}^{(3)}\right)G_j\right|^p\right)\right\}^{1\vee p/2} \leq n^{-p+1\vee\frac{p}{2}}b_n^{-p(1-\alpha)}.$$

And with A2, (2.20), (2.21) and Lemma A.1 (a. and b.) we get:

$$\mathbb{E}\left(\left|Q_n^{(3)}\right|^p\right) \leq Cn^{-\alpha p}b_n^{-p(1-\alpha)}.$$

Again, (2.4) and Lemma A.1 imply that

$$\mathbb{E}\left(\left|Z_{n,j}^{(4)}\right|^{2p}\right) \leq C\left\{\sum_{\ell=0}^{j-1}|A_{n,\ell}|\right\}^{2p} \leq C|b_n|^{2p}, \tag{2.22}$$

which implies that

$$\mathbb{E}\left(\left|Q_n^{(4)}\right|^p\right) \leq Cb_n^p.$$

In conclusion

$$\mathbb{E}\left(\left|\sum_{i=1}^n Q_n^{(i)}\right|^p\right) \leq C\left(b_n + n^{-\alpha}b_n^{-1+\alpha} + n^{-1+\frac{1}{2}\vee\frac{1}{p}}b_n^{-1/2}\right)^p,$$

and the theorem easily follows. □

2.2. *Proof of Theorem 1.3.*

Proof. Again we show that under the stated assumptions, each terms on the right-hand side of (2.5) converges almost surely to zero. Clearly, since $\sum_n(n^{-1}b_n^{-1})^p < \infty$ the term ϵ_n poses no problem.

`On term` $D_n^{(1)}$. We have seen in the proof of Theorem 1.1 that

$$\mathbb{E}\left(\left|\sum_{j=1}^m G_j P_{\theta_{j-1}}g_{\theta_{j-1}}(X_{j-1})\right|^p\right) \leq Cm^{1\vee(p/2)}.$$

Furthermore, $n^{-1}|1-w(b_n)| \leq c_w n^{-1}b_n$. This is enough to conclude (using Proposition A.1), that $D_n^{(1)}$ converges almost surely to zero.

`On terms` $D_n^{(2)}$, $D_n^{(3)}$, $D_n^{(4)}$, $D_n^{(5)}$. We have

$$\left|D_n^{(2)}\right| \leq Cn^{-1}\sum_{k=1}^{n}\gamma_k V^{\eta+\beta}(X_k)\left(V^\beta(X_k)+V^\beta(X_{k-1})\right). \tag{2.23}$$

and

$$\mathbb{E}\left(\sum_{k\geq 1}k^{-1}\gamma_k V^{\eta+\beta}(X_k)\left(V^\beta(X_k)+V^\beta(X_{k-1})\right)\right) \leq C\sum_{k\geq 1}k^{-1}\gamma_k < \infty.$$

By Kronecker lemma the right-hand size of (2.23) converges almost surely to zero and so is its left-hand side. Similar arguments apply to $D_n^{(3)}$, $D_n^{(4)}$ and $D_n^{(5)}$. We omit these details.

`On terms` $Q_n^{(1)}$. We recall that $Q_n^{(1)} = \sum_{j=2}^{n} D_{n,j}$ with $D_{n,j} = n^{-1}Z_{n,j}^{(1)}G_j$. Borrowing the bound (2.19) we get:

$$\sup_{j\geq 2}\mathbb{E}^{1/p}\left(|D_{n,j}|^p\right) \leq Cn^{-1}\sup_{j\geq 2}\mathbb{E}^{1/2p}\left(|Z_{n,j}^{(1)}|^{2p}\right) \leq Cn^{-1}b_n^{-1/2}.$$

$D_{n,j}-D_{n-1,j} = \left(n^{-1}Z_{n,j}^{(1)} - (n-1)^{-1}Z_{n-1,j}^{(1)}\right)G_j$ and

$$n^{-1}Z_{n,j}^{(1)} - (n-1)^{-1}Z_{n-1,j}^{(1)} = \sum_{\ell=1}^{j-1}\left(n^{-1}w\left((j-\ell)b_n\right) - (n-1)^{-1}w\left((j-\ell)b_{n-1}\right)\right)G_\ell,$$

which again forms a martingale for each n and we deduce using (2.1) that

$$\begin{aligned}&\mathbb{E}\left(\left|n^{-1}Z_{n,j}^{(1)} - (n-1)^{-1}Z_{n-1,j}^{(1)}\right|^{2p}\right)\\ &\qquad \leq C\left\{\sum_{\ell=1}^{j-1}\left|n^{-1}w\left(\ell b_n\right) - (n-1)^{-1}w\left(\ell b_{n-1}\right)\right|^2\right\}^p \leq Cn^{-4p}b_n^{-p}.\end{aligned}$$

So that

$$\sup_{j\geq 2}\mathbb{E}^{1/p}\left(|D_{n,j}-D_{n-1,j}|^p\right) \leq Cn^{-2}b_n^{-1/2}.$$

By assumption, $\sum_{n\geq 1}\left(n^{-1}b_n^{-1/2}\right)^{p\wedge 2} < \infty$ and $\sum_{n\geq 1} n^{\frac{1}{2}\vee\frac{1}{p}}n^{-2}b_n^{-1/2} < \infty$, and we use Theorem A.1 to conclude that $Q_n^{(1)}$ converges almost surely to zero.

`On term` $Q_n^{(2)}$. $Q_n^{(2)} = \sum_{j=2}^n D_{n,j}$, where $D_{n,j} = 4n^{-1}(Z_{n,j}^{(2)} + Z_{n,j}^{(3)})G_j$. Again, we shall apply Theorem A.1. (2.20) and (2.21) yield

$$\sup_{j\geq 2} \mathbb{E}^{1/p}\left(|D_{n,j}|^p\right) \leq C\mathbb{E}^{1/(2p)}\left(\left|n^{-1}\left(Z_{n,j}^{(2)} + Z_{n,j}^{(3)}\right)\right|^{2p}\right) \leq Cn^{-1}b_n^{-(1-\alpha)}$$

In a similar way, we calculate that

$$\begin{aligned}
&\mathbb{E}^{1/p}\left(|D_{n,j} - D_{n-1,j}|^p\right) \\
&\quad \leq C\mathbb{E}^{1/2p}\left(\left|n^{-1}\left(Z_{n,j}^{(2)} + Z_{n,j}^{(3)}\right) - (n-1)^{-1}\left(Z_{n-1,j}^{(2)} + Z_{n-1,j}^{(3)}\right)\right|^{2p}\right) \\
&\qquad\qquad \leq C(n-1)^{-2}\mathbb{E}^{1/2p}\left(\left|\left(Z_{n,j}^{(2)} + Z_{n,j}^{(3)}\right)\right|^{2p}\right) \\
&\qquad + C(n-1)^{-1}\mathbb{E}^{1/2p}\left(\left|\left(Z_{n,j}^{(2)} + Z_{n,j}^{(3)}\right) - \left(Z_{n-1,j}^{(2)} + Z_{n-1,j}^{(3)}\right)\right|^{2p}\right) \\
&\leq C\left(n^{-3}b_n^{-(1-\alpha)} + n^{-1}\left(n^{-1} + \sum_{\ell=1}^{j-1}|w((j-\ell)b_n) - w((j-\ell)b_{n-1})|\gamma_\ell\right)\right) \\
&\qquad\qquad\qquad\qquad \leq Cn^{-2}b_n^{-(1-\alpha)}.
\end{aligned}$$

And in view of (1.8), we conclude that $Q_n^{(2)}$ converges almost surely to zero.

`On terms` $Q_n^{(3)}$`: .` We shall apply Toeplitz Lemma. For any $\delta > 0$,

$$\begin{aligned}
&\sum_{n\geq 1}\mathbb{P}\left(|g_{\theta_{n-1}}(X_n) - g_{\theta_n}(X_n)| > \delta\right) \\
&\qquad\qquad \leq C\delta^{-2}\sum_{n\geq 1}\gamma_n^2\mathbb{E}\left(V^{2(\beta+\eta)}(X_n)\right) \leq C\delta^{-2}\sum_{n\geq 1}\gamma_n^2,
\end{aligned}$$

thus $|g_{\theta_{n-1}}(X_n) - g_{\theta_n}(X_n)|$ converges almost surely to zero as $n \to \infty$. Also, it is clear that for any $j \geq 2$ fixed, $n^{-1}\left(2Z_{n,j}^{(2)} + Z_{n,j}^{(3)}\right)$ converges almost surely to zero as $n \to \infty$, since w is bounded. By Toeplitz lemma, in order to conclude that $Q_n^{(3)}$ converges almost surely to zero, it suffices to show that

$$\sup_{n\geq 2}\sum_{j=2}^n n^{-1}\left|2Z_{n,j}^{(2)} + Z_{n,j}^{(3)}\right| < \infty, \quad \text{almost surely.}$$

$$\begin{aligned}
\sum_{j=2}^n n^{-1}\left|Z_{n,j}^{(3)}\right| &\leq n^{-1}\sum_{\ell=1}^{n-1}\sum_{j=\ell+1}^n |w((j-\ell)b_n)|\left|g_{\theta_{\ell-1}}(X_\ell) - g_{\theta_\ell}(X_\ell)\right| \\
&\qquad\qquad\qquad \leq Cn^{-1}\sum_{\ell=1}^{n-1}\gamma_\ell V^{\eta+\beta}(X_\ell).
\end{aligned}$$

And it is easy to see that the term on the rhs converges almost surely to zero. Similarly

$$n^{-1}\sum_{j=2}^{n}\left|Z_{n,j}^{(2)}\right| \leq Cn^{-1}\sum_{\ell=1}^{n-1}V^{\beta}(X_{\ell-1})\sum_{j=\ell+1}^{n}\left|w\left((j-\ell)b_n\right)-w\left((j-\ell+1)b_n\right)\right|$$
$$\leq Cn^{-1}\sum_{\ell=1}^{n-1}V^{\beta}(X_{\ell-1}).$$

The last inequality uses the bound (2.4). By the strong law of large of Proposition 2.1 of [2], $n^{-1}\sum_{k=1}^{n}V^{\beta}(X_k)$ converges almost to $\pi(V^{\beta})<\infty$ and is therefore bounded almost surely.

`On terms` $Q_n^{(4)}$. $Q_n^{(4)}=n^{-1}\sum_{j=1}^{n-1}Z_{n,j}^{(4)}P_{\theta_j}g_{\theta_j}(X_j)$, where $Z_{n,j}^{(4)}$ is given by $Z_{n,j}^{(4)}=\sum_{\ell=0}^{j-1}A_{n,j-\ell}P_{\theta_\ell}g_{\theta_\ell}(X_\ell)$. To deal with $Q_n^{(4)}$, we need the divided difference sequences introduced in Appendix B: $\Delta_n^{(0)}(\ell)=w(\ell b_n)$, and for $q\geq 1$

$$\Delta_{n,q,\ell}=\Delta_n^{(q)}\left(\ell-q,\ldots,\ell\right)$$
$$\stackrel{\text{def}}{=}\frac{b_n^{-1}}{q}\left(\Delta_n^{(q-1)}\left(\ell-q+1,\ldots,\ell\right)-\Delta_n^{(q-1)}\left(\ell-q,\ldots,\ell-1\right)\right).$$

In particular, $A_{n,\ell}=-2b_n^2\Delta_{n,2,\ell+1}$ so that $Q_n^{(4)}$ becomes

$$Q_n^{(4)}=-2n^{-1}b_n^2\sum_{j=1}^{n-1}P_{\theta_j}g_{\theta_j}(X_j)\left\{\sum_{\ell=0}^{j-1}\Delta_{n,2,j-\ell+1}P_{\theta_\ell}g_{\theta_\ell}(X_\ell)\right\}.$$

We will show below that there exists a family $\{f_\theta\in\mathcal{L}_{V^\beta},\ \theta\in\Theta\}$ of measurable functions for which $\sup_{\theta\in\Theta}|f_\theta|_{V^\beta}<\infty$ and $|f_\theta-f_{\theta'}|_{V^\beta}\leq C|h|_{V^\beta}D_\beta(\theta,\theta')$ for some finite constant C, such that

$$Q_n^{(4)} = -3!n^{-1}b_n^3\sum_{j=1}^{n-1}f_{\theta_j}(X_j)\left\{\sum_{\ell=0}^{j-1}\Delta_{n,3,j-\ell+2}P_{\theta_\ell}g_{\theta_\ell}(X_\ell)\right\}+\epsilon_n^{(3)}, \tag{2.24}$$

where the sequence $\epsilon_n^{(3)}$ converges almost surely to zero. By iterating the same argument on the first term on the rhs of (2.24) and for r as in the statement of the theorem, it follows that there exists a family $\{f_\theta^{(r)}\in\mathcal{L}_{V^\beta},\ \theta\in\Theta\}$ of measurable functions for which $\sup_{\theta\in\Theta}|f_\theta^{(r)}|_{V^\beta}<\infty$ and $|f_\theta^{(r)}-f_{\theta'}^{(r)}|_{V^\beta}\leq C|h|_{V^\beta}D_\beta(\theta,\theta')$ for some finite constant C, such that

$$Q_n^{(4)} = -r!n^{-1}b_n^r\sum_{j=1}^{n-1}f_{\theta_j}^{(r)}(X_j)\left\{\sum_{\ell=0}^{j-1}\Delta_{n,r,j-\ell+r-1}P_{\theta_\ell}g_{\theta_\ell}(X_\ell)\right\}+\epsilon_n^{(r)},$$

where $\epsilon_n^{(r)}$ converges almost surely to zero. Moreover, we can easily calculate that

$$\mathbb{E}\left(\left|n^{-1}b_n^r\sum_{j=1}^{n-1}f_{\theta_j}^{(r)}(X_j)\left\{\sum_{\ell=0}^{j-1}\Delta_{n,r,j-\ell+r-1}P_{\theta_\ell}g_{\theta_\ell}(X_\ell)\right\}\right|^p\right) \leq Cb_n^{(r-1)p},$$

and since $\sum b_n^{(r-1)p}<\infty$, we then conclude that $Q_n^{(4)}$ converges almost surely to zero. We now prove the claim (2.24).

We introduce the function f_θ defined as

$$f_\theta(x)=\sum_{\ell\geq 0}P_\theta^\ell\left(P_\theta g_\theta\right)(x),\ \ x\in\mathsf{X},\ \theta\in\Theta.$$

f_θ is well defined and belongs to $\mathcal{L}_{V^\beta}$. Moreover, as in Lemma A.1 (a)-(b), there exists a finite constant C such that $\sup_{\theta\in\Theta}|f_\theta|_{V^\beta}<\infty$ and $|f_\theta-f_{\theta'}|_{V^\beta}\leq C|h|_{V^\beta}D_\beta(\theta,\theta')$. Using f_θ we then rewrite $Q_n^{(4)}$ as follows.

$$\begin{aligned}Q_n^{(4)}=n^{-1}&\sum_{j=2}^{n}Z_{n,j}^{(4)}\left(P_{\theta_j}g_{\theta_j}(X_j)-P_{\theta_{j-1}}g_{\theta_{j-1}}(X_j)\right)\\&+n^{-1}\sum_{j=2}^{n}Z_{n,j}^{(4)}\left(f_{\theta_{j-1}}(X_j)-P_{\theta_{j-1}}f_{\theta_{j-1}}(X_{j-1})\right)\\&+n^{-1}\sum_{j=2}^{n}\left(Z_{n,j}^{(4)}-Z_{n,j-1}^{(4)}\right)P_{\theta_{j-1}}f_{\theta_{j-1}}(X_{j-1})\\&+n^{-1}\sum_{j=2}^{n}Z_{n,j-1}^{(4)}P_{\theta_{j-1}}f_{\theta_{j-1}}(X_{j-1})-Z_{n,j}^{(4)}P_{\theta_j}f_{\theta_j}(X_j)\\+n^{-1}\sum_{j=2}^{n}Z_{n,j}^{(4)}&\left(P_{\theta_j}f_{\theta_j}(X_j)-P_{\theta_{j-1}}f_{\theta_{j-1}}(X_j)\right)=U_n^{(1)}+U_n^{(2)}+U_n^{(3)}+U_n^{(4)}+U_n^{(5)},\end{aligned}$$

say. The term $U_n^{(3)}$ can be written as

$$\begin{aligned}U_n^{(3)}=n^{-1}A_{n,1}&\sum_{j=1}^{n-1}P_{\theta_j}f_{\theta_j}(X_j)P_{\theta_j}g_{\theta_j}(X_j)\\&-6b_n^3\sum_{j=1}^{n-1}P_{\theta_j}f_{\theta_j}(X_j)\sum_{\ell=0}^{j-1}\Delta_{n,3,j-\ell+2}P_{\theta_\ell}g_{\theta_\ell}(X_\ell).\end{aligned}$$

By the strong law of large numbers and the fact that $|A_{n,1}|\leq cb_n^2$, the first term

$n^{-1}A_{n,1}\sum_{j=1}^{n-1} P_{\theta_j} f_{\theta_j}(X_j) P_{\theta_j} g_{\theta_j}(X_j)$ converges almost surely to zero, while the second term in $U_n^{(3)}$ provides the leading term in (2.24).

The terms $U_n^{(1)}$ and $U_n^{(5)}$ are similar to $Q_n^{(3)}$ and can be shown to converge to zero with the same Toeplitz lemma argument. The key point is to show that $\sup_{n\geq 2} n^{-1}\sum_{j=2}^{n} |Z_{n,j}^{(4)}| < \infty$ almost surely. But

$$n^{-1}\sum_{j=2}^{n}\left|Z_{n,j}^{(4)}\right| \leq Cn^{-1}\sum_{\ell=1}^{n-1} V^{\beta}(X_\ell)\left(\sum_{\ell=0}^{j-1}|A_{n,j-\ell}|\right) \leq Cn^{-1}b_n\sum_{\ell=0}^{n} V^{\beta}(X_\ell),$$

where the last inequality uses Proposition B.1.

The term $U_n^{(2)}$ is a martingale array that we write as $U_n^{(2)} = \sum_{j=2}^{n} D_{n,j}$, where $D_{n,j} = n^{-1}Z_{n,j}^{(4)}\left(f_{\theta_{j-1}}(X_j) - P_{\theta_{j-1}} f_{\theta_{j-1}}(X_{j-1})\right)$. We handle this term as $Q_n^{(1)}$ and $Q_n^{(2)}$ above, using Theorem A.1. We calculate easily that

$$\sup_{j\geq 1}\mathbb{E}^{1/p}\left(|D_{n,j}|^p\right) \leq Cn^{-1}b_n.$$

and

$$\mathbb{E}^{1/p}\left(|D_{n,j} - D_{n-1,j}|^p\right) \leq C\left(n^{-2}b_n + n^{-1}\sum_{\ell=0}^{j-1}|A_{n,j-\ell} - A_{n-1,j-\ell}|\right).$$

By the Mean Value theorem for divided differences ([4] Theorem 2.9),

$$\begin{aligned}
&|A_{n,j-\ell} - A_{n-1,j-\ell}| = 2\left|b_n^2\Delta_{n,2,j-\ell+1} - b_{n-1}^2\Delta_{n-1,2,j-\ell+1}\right| \\
&\leq C\left(n^{-1}b_n^2 + b_n^2\int_0^1 dt_1\int_0^{t_1} dt_2\left|w^{(2)}\left(b_n(t_1+t_2)\right) - w^{(2)}\left(b_{n-1}(t_1+t_2)\right)\right|\right) \\
&\qquad\leq Cn^{-1}b_n^2.
\end{aligned}$$

We conclude that

$$\sup_{j\geq 1}\mathbb{E}^{1/p}\left(|D_{n,j} - D_{n-1,j}|^p\right) \leq Cn^{-2}b_n.$$

Given the assumptions and Theorem A.1, $U_n^{(2)}$ also converges almost surely to zero.

For the term $U_n^{(4)}$, we have $U_n^{(4)} = n^{-1}\left(Z_{n,1}^{(4)}P_{\theta_1}f_{\theta_1}(X_1) - Z_{n,n}^{(4)}P_{\theta_n}f_{\theta_n}(X_n)\right)$ which has a p-moment bounded by $Cn^{-p}b_n^{-p}$ and since $\sum n^{-p}b_n^{-p} < \infty$, we conclude easily that $U_n^{(4)}$ converges almost surely to zero.

□

2.3. *Proof of Theorem 1.2.*

PROOF. With $\beta+\eta<1/4$, we can set $p=\frac{1}{2(\beta+\eta)}>2$. With this choice and $b_n=n^{-\delta}$, $\delta\in(2(\beta+\eta),1/2)$ all the assumptions in Theorem 1.3 hold true except the summability assumption $\sum_n b_n^{(r-1)p}<\infty$ and the fact that w admits continuous derivatives of order greater that 2. One can check by inspecting the proof of Theorem 1.3 that these latter assumptions are only used in dealing with the term $Q_n^{(4)}$. Thus, to prove Theorem 1.2, it suffices to show that $Q_n^{(4)}$ converges almost surely to zero when $\beta+\eta<1/4$.

But from the proof of Theorem 1.1, we have obtained the bound $\mathbb{E}\left(\left|Q_n^{(4)}\right|^p\right)\leq Cb_n^p=O(n^{-p\delta})$, and we have $p\delta>1$. The theorem then follows. $\square$

APPENDIX A: SOME LIMIT THEOREMS FOR MARTINGALE ARRAYS

We recall the following lemma proved in [2].

LEMMA A.1. *Assume that $\{P_\theta,\ \theta\in\Theta\}$ satisfies A1.*

a. *There exists a finite constant C such that*

$$\sup_{n\geq 0}\mathbb{E}\left(V(X_n)\right)\leq C. \tag{A.1}$$

b. *Let $\beta\in(0,1]$ and $\{h_\theta\in\mathcal{L}_{V^\beta},\ \theta\in\Theta\}$ be such that $\pi(h_\theta)=0$, $\sup_{\theta\in\Theta}|h_\theta|_{V^\beta}<\infty$. Then the function $\tilde g_\theta\stackrel{\text{def}}{=}\sum_{j\geq 0}P_\theta^j h_\theta(x)$ is well-defined, $|\tilde g_\theta|_{V^\beta}\leq C|h_\theta|_{V^\beta}$, where the constant C does not depend on $\{h_\theta\in\mathcal{L}_{V^\beta},\ \theta\in\Theta\}$. Moreover we can take C such that for any $\theta,\theta'\in\Theta$,*

$$|\tilde g_\theta-\tilde g_{\theta'}|_{V^\beta}\leq C\sup_{\theta\in\Theta}|h_\theta|_{V^\beta}\left(D_\beta(\theta,\theta')+|h_\theta-h_{\theta'}|_{V^\beta}\right). \tag{A.2}$$

c. *Assume A2. Let $\beta\in(0,1-\eta)$ and $h\in\mathcal{L}_{V^\beta}$ be such that $\pi(h)=0$. Define $S_n(j)=\sum_{\ell=j+1}^{j+n}h(X_\ell)$. Let $p\in(1,(\beta+\eta)^{-1})$. Then there exists a finite constant C that does not depend on n,j,θ nor h such that*

$$\mathbb{E}\left(|S_n(j)|^p\right)\leq C|h|_{V^\beta}n^{1\vee(p/2)}.$$

We shall need the following classical law of large numbers for martingales.

PROPOSITION A.1. *Let $\{(S_k,\mathcal{F}_k),\ k\geq 1\}$ be a martingale such that*

$$\sum_{n\geq 1}n^{-p-1}\mathbb{E}\left(|S_n|^p\right)<\infty,$$

for some $p>1$. Then $n^{-1}S_n$ converges almost surely to zero.

We now give an extension of this result to martingale arrays. Let $\{D_{n,i}, \mathcal{F}_i,\ 1 \leq i \leq n\}$, $n \geq 1$ be a martingale-difference array. That is $\{\mathcal{F}_i,\ i \geq 0\}$ is a non-decreasing sequence of sigma-algebra, for any $1 \leq i \leq n$, $\mathbb{E}\left(|D_{n,i}|\right) < \infty$ and $\mathbb{E}\left(D_{n,i}|\mathcal{F}_{i-1}\right) = 0$. For $p > 1$, we introduce the quantities

$$v_n(p) \stackrel{\text{def}}{=} \sup_{j \geq 1} \mathbb{E}^{1/p}\left(|D_{n,j}|^p\right), \quad \text{and} \quad \Lambda_n(p) = \sup_{j \geq 1} \mathbb{E}^{1/p}\left(|D_{n,j} - D_{n-1,j}|^p\right).$$

THEOREM A.1. *Let $\{D_{n,i}, \mathcal{F}_i,\ 1 \leq i \leq n\}$, $n \geq 1$ be a martingale-difference array. Suppose that there exists $p > 1$ such that*

$$\sum_{n \geq 1} \left(v_n(p)\right)^{p \wedge 2} < \infty, \quad \textit{and} \quad \sum_{n \geq 1} n^{\frac{1}{2} \vee \frac{1}{p}} \Lambda_n(p) < \infty.$$

Then $\sum_{i=1}^n D_{n,i}$ converges almost surely to zero.

PROOF. We introduce the partial sums $M_{n,k} := \sum_{j=1}^k D_{n,j}$, $1 \leq k \leq n$ and $S_{n,k} = M_{n,k} + \sum_{j=n+1}^k D_{j,j}$, $k \geq 1$, where $\sum_a^b \cdot = 0$ if $a > b$. We write M_n instead of $M_{n,n}$. Clearly $\{S_{n,k}, \mathcal{F}_k,\ k \geq 1\}$ is a martingale for each n. We also introduce $R_n := \sum_{j=1}^{n-1}\left(D_{n,j} - D_{n-1,j}\right)$, $n \geq 2$ ($R_1 = 0$). For $m \geq n$, $M_m = S_{n,m} + \sum_{j=n+1}^m R_j$. Thus, for $k \geq 1$,

$$\max_{n \leq m \leq n+k} |M_m| \leq \max_{n \leq m \leq n+k} |S_{n,m}| + \sum_{j=n+1}^{n+k} |R_j|.$$

Therefore for any $\lambda > 0$,

(A.3)
$$\begin{aligned}
\mathbb{P}\left(\max_{n \leq m \leq n+k} |M_m| > \lambda\right) &\leq \mathbb{P}\left(\max_{n \leq m \leq n+k} |S_{n,m}| > \lambda/2\right) + \mathbb{P}\left(\sum_{j=n+1}^{n+k} |R_j| > \lambda/2\right) \\
&\leq 2^{-p}\lambda^p \mathbb{E}\left(|S_{n,n+k}|^p\right) + 2^{-p}\lambda^p \left(\sum_{j=n+1}^{\infty} \mathbb{E}^{1/p}\left(|R_j|^p\right)\right)^p.
\end{aligned}$$

The last inequality uses Doob's inequality for martingales and Markov's and Minkowski's inequalities.

By (2.1) we have:

(A.4)

$$\mathbb{E}\left(|S_{n,n+k}|^p\right) \le C\left(\sum_{j=1}^{n}\mathbb{E}^{1\wedge(2/p)}\left(|D_{n,j}|^p\right)+\sum_{j=n+1}^{n+k}\mathbb{E}^{1\wedge(2/p)}\left(|D_{j,j}|^p\right)\right)^{1\vee(p/2)}$$
$$\le C\left(n\left(v_n(p)\right)^{p\wedge 2}+\sum_{j=n+1}^{\infty}\left(v_n(p)\right)^{p\wedge 2}\right)^{1\vee(p/2)},$$

and the expression on the rhs converges to zero as $n\to\infty$, by assumption. Similarly, notice that $\{\sum_{j=1}^{k}\left(D_{n,j}-D_{n-1,j}\right),\ 1\le k\le n-1\}$ is a martingale wrt $\{\mathcal{F}_k,\ k\ge 1\}$ and as above we get

(A.5)

$$\mathbb{E}^{1/p}\left(|R_n|^p\right)\le C\left\{\sum_{\ell=1}^{n-1}\mathbb{E}^{1\wedge(2/p)}\left(|D_{n,\ell}-D_{n-1,\ell}|^p\right)\right\}^{(1/2)\vee(1/p)} \le Cn^{(1/2)\vee(1/p)}\Lambda_n(p).$$

Combining (A.3), (A.4) and (A.5), we conclude that

$$\lim_{n\to\infty}\mathbb{P}\left(\sup_{m\ge n}|M_m|>\lambda\right)=0,$$

under the stated assumptions. □

APPENDIX B: A BOUND ON DIVIDED DIFFERENCE SEQUENCES

Define $\Delta_n^{(0)}(\ell) = w(\ell b_n)$, $\ell\in\{0,\ldots,b_n^{-1}\}$ and for $p\ge 1$, and $\ell\in\{p,\ldots,b_n^{-1}\}$,

$$\Delta_n^{(p)}\left(\ell-p,\ldots,\ell\right)=\frac{b_n^{-1}}{p}\left(\Delta_n^{(p-1)}\left(\ell-p+1,\ldots,\ell\right)-\Delta_n^{(p-1)}\left(\ell-p,\ldots,\ell-1\right)\right).$$

In particular, $\Delta_n^{(1)}(\ell-1,\ell) = b_n^{-1}\left(w(\ell b_n)-w((\ell-1)b_n)\right)$, $\Delta_n^{(2)}(\ell-2,\ell-1,\ell) = -2^{-1}b_n^{-2}A_{n,\ell-1}$, and $\Delta_n^{(3)}(\ell-3,\ell-2,\ell-1,\ell)=6b_n^{-3}\left(A_{n,\ell-1}-A_{n,\ell-2}\right)$, where $A_{n,\ell}=2w\left(\ell b_n\right)-w\left((\ell-1)b_n\right)-w\left((\ell+1)b_n\right)$. If w is p times continuous differentiable on $[0,1]$ and we denote by $w^{(p)}$ its p-th derivative, the Mean Value Theorem for divided differences (see e.g. [4], Theorem 2.10) asserts that there exists $\theta_{n,\ell}\in[(\ell-p)b_n,\ell b_n]$ such that

$$\Delta_n^{(p)}\left(\ell-p,\ldots,\ell\right)=\frac{w^{(p)}(\theta_{n,\ell})}{p\,!}.$$

Therefore there exists a finite constant c_p such that for all $n \geq 1$, $\ell \in \{p, \ldots, b_n^{-1}\}$,

$$\left|\Delta_n^{(p)}\left(\ell - p, \ldots, \ell\right)\right| \leq c_p.$$

In particular we have:

PROPOSITION B.1. *Assume that A3 hold. Then there exists a finite constant c such that for any $n \geq 1$,*

$$\sum_{\ell=-n}^{n} |w(\ell b_n)| \leq c b_n^{-1}, \quad \sum_{\ell=-n}^{n} |w((\ell-1)b_n) - w(\ell b_n)| \leq c, \quad \sum_{\ell=-n}^{n} |A_{n,\ell-1}| \leq c b_n,$$

$$\text{and} \quad \sum_{\ell=-n}^{n} |A_{n,\ell-2} - A_{n,\ell-1}| \leq c b_n^2.$$

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